Lagrangian Variable Cosmology: Theoretical Reconstruction of the Lock-φ Modulation under a Phase-Rotational Spacetime-Unfolding Hypothesis

Working Paper

LUMENPIXEL
Independent researcher, Busan, Republic of Korea
Computational assistance: Claude (Anthropic)
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Abstract

The author has previously developed a phenomenological line of work, LVC v1–v16, passing through several structural revisions and identifying a localized late-time feature in the cosmological expansion rate, fitted on a combined BAO + supernovae + distance-prior dataset at N=1738. The most recent identification, Lock-φ, locates a Gaussian feature at the natural-constant position $z_c=1/\varphi$ with width $w=e/(5\pi)$. The construction of the present paper proceeds from a working hypothesis that spacetime unfolds through the rotational expansion of a compact phase variable; this hypothesis is stated as the starting point of the construction and is to be addressed in future work. The present working paper marks a transition from that phenomenology to a theoretical construction: a cosmology derived from a parametric pendulum equation on the logarithmic-time variable $\theta=\ln(1+z)$, in which the expansion history is the sum of the matter sector and a derived dark-energy density $\Omega_{\mathrm{LVC}}(z)$, with no separate cosmological-constant term in the Friedmann equation. The construction introduces four axioms and produces, by direct integration, a Gaussian profile in θ. Three of the four model parameters — the critical point $\theta_c=\ln\varphi$, the coupling width $w=e/(5\pi)$, and the coupling scale $\Omega_0=\sqrt{2}\cdot 5\pi/e$ — are locked to natural-constant combinations a priori; only the boost amplitude A is fitted. The derived $\Omega_{\mathrm{LVC}}(z)$ is not constant: it departs from its present-day value by up to about 21% near the critical point and returns to its asymptotic value at high z. The corresponding effective equation of state crosses $w=-1$. On the joint Pantheon+ unbinned, DESI DR1+DR2 BAO, BOSS, eBOSS, Planck distance priors, and SH0ES dataset (N=1738) the construction yields χ^2=1602.77 against a ΛCDM χ^2=1624.85, corresponding to ΔBIC=−14.62 at k=4. The previously published phenomenological Lock-φ-A fit on the same dataset achieves χ^2=1601.19, a difference of $\Delta\chi^2$=+1.58 at the same parameter count. ΛCDM emerges as the A=0 special case of the construction; for nonzero A the dark-energy density is dynamical.

1. Introduction

The phenomenological model line LVC v1–v16 has converged, over many revisions, on a single observational claim: the joint dataset of Pantheon+ supernovae, DESI BAO, Planck distance priors, and SH0ES H_0 is fitted with a meaningful improvement over ΛCDM by a multiplicative factor $T(z)$ that modulates the expansion rate, with a Gaussian-localized feature centered on a redshift consistent with $z_c=1/\varphi$ and width consistent with $w=e/(5\pi)$; in the A=0 limit this factor reduces to unity and the expansion history coincides with ΛCDM. This identification, labeled Lock-φ in v16, has ΔBIC=−16.20 against ΛCDM at k=4 on N=1738 measurements. The choice of a Gaussian functional form, however, has been to date a phenomenological convenience: no construction has been offered for why the boost takes that profile, and no derivation has been given for why the position and width should be tied to φ and to the combination $e/(5\pi)$ respectively.

The present working paper offers a construction. The Lock-φ modulation is interpreted as the closed-form solution of a parametric harmonic oscillator on the logarithmic-time variable θ=ln(1+z). The construction introduces four axioms and produces the Gaussian as the unique solution with two specified initial conditions at the critical point. Three of the four model parameters are fixed by natural-constant combinations and are not adjusted to the data; the fourth, the boost amplitude, remains a continuous parameter and is fitted.

The construction proceeds from a single working hypothesis: that spacetime unfolds through the rotational expansion of a compact phase variable, whose dynamics on the logarithmic time coordinate take the form of a parametric pendulum equation. This hypothesis is not derived from first principles in the present paper; it is stated as the starting point of the construction and is to be addressed in future work. The discipline of the present paper is to develop the consequences of the hypothesis as a closed-form expansion history and to compare it with data, without claiming a physical interpretation of the rotation itself beyond this framing.

The Friedmann equation is written without a separate cosmological-constant term: the expansion history is $H^2(z)=H_0{}^2[\Omega_m(1+z)^3+\Omega_{\mathrm{LVC}}(z)]$, where the dimensionless dark-energy density $\Omega_{\mathrm{LVC}}(z)$ is derived from the modulation factor and is not constant. Consequently the construction is not a model in which a constant Λ is assumed and a small modulation added on top; it is a model in which a single time-dependent dark-energy density takes the place of Λ. The ΛCDM expansion history corresponds to the A=0 special case of the construction, in which $\Omega_{\mathrm{LVC}}(z)$ reduces to the constant value $(1-\Omega_m)$. The derived form fits the same dataset slightly less well than the original phenomenological form on a free z-Gaussian — $\Delta\chi^2=+1.58$ at $k=4$ — while reducing the modulation parameters from one position, one width, and one amplitude to one amplitude alone. The construction may fail in subsequent tests; the discipline of the present paper is to state it as it stands.

2. Axioms

The construction rests on four axioms. The axioms are stated as starting assumptions and are not derived in this paper. The matter sector is taken as a pressureless dust scaling as $(1+z)^3$, in agreement with standard cosmology; the dynamical content of the construction is confined to the dark-energy density. The pendulum form of Axiom L2 below reflects the phase-rotational spacetime-unfolding hypothesis stated in Section1: a compact phase variable, whose rotational expansion is taken to underlie the unfolding of cosmological spacetime, oscillates on the logarithmic time θ and produces a multiplicative deviation from the A=0 background at the level of the expansion rate. The construction does not introduce the rotation as an additional dynamical field; it represents its observable signature only.

Axiom L1 (Logarithmic time).

The dynamical time variable of the construction is $\theta\equiv\ln(1+z)$. It runs from $\theta=0$ at the present epoch to large positive values toward the early universe. In a de Sitter expansion $a\propto\exp(H_0t)$, θ is $-\ln a$ and is therefore proportional to H_0t; in matter domination it is approximately $(2/3)\ln(t/t_0)$. Within the construction it is the time variable along which the modulation factor evolves; no claim about a globally preferred frame is made.

Axiom L2 (Phase pendulum equation).

The deviation amplitude $v(\theta)\equiv T(\theta)-1$, where $T(\theta)$ is the multiplicative modulation factor relating the expansion rate to its A=0 limit, satisfies a homogeneous linear second-order ordinary differential equation with a time-dependent coupling:

$$v''(\theta)+\Omega^2(\theta)\cdot v(\theta)=0$$

Primes denote derivatives with respect to θ. No source term is admitted on the right-hand side; the dynamics of the deviation are autonomous in θ. The form of $\Omega^2(\theta)$ is fixed by Axioms L3 and L4 below.

Axiom L3 (Critical point at the golden ratio).

The coupling $\Omega^2(\theta)$ attains its maximum at the position

$$\theta_c=\ln\varphi, \varphi=(1+\sqrt{5})/2$$

Equivalently, $1+z_c=\varphi$, so that $z_c=1/\varphi\approx0.618$ and the scale-factor at the critical point is $a_c=1/\varphi$. The position is taken from the v16 Lock-φ phenomenological identification.

Axiom L4 (Coupling profile and width).

The coupling has the parabolic form

$$\Omega^2(\theta)=(2/w^2)\cdot[1-2((\theta-\theta_c)/w)^2],$$

with the width w locked to

$$w=e/(5\pi)\approx0.173.$$

The coupling is positive (bound) for $|\theta-\theta_c|<w/\sqrt{2}$ and negative (unbound) outside that range. The maximal value attained at the critical point is $\Omega_0{}^2=2/w^2=50\pi^2/e^2$, yielding $\Omega_0=\sqrt{2}\cdot5\pi/e\approx8.17$.

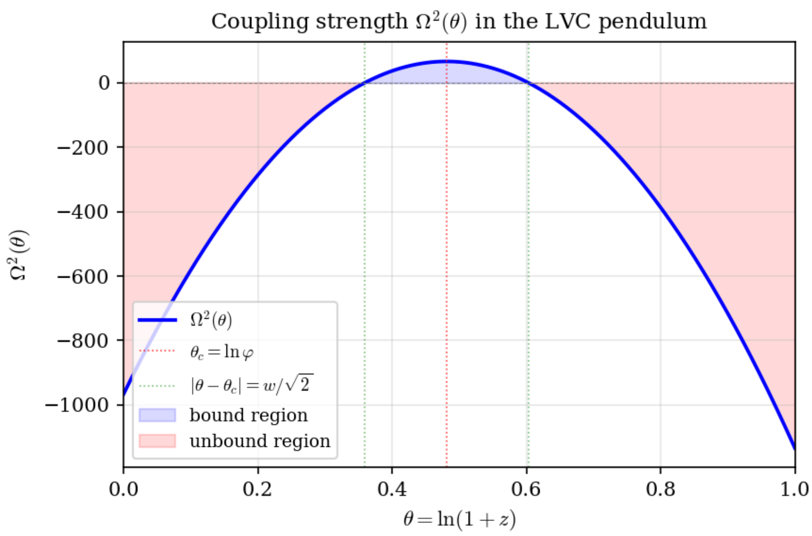

Figure 1. The coupling profile $\Omega^2(\theta)$ defined by Axiom L4. The maximum is attained at $\theta_c=\ln\varphi$ (red dotted line) with $\Omega_0{}^2=2/w^2$. The coupling vanishes at $|\theta-\theta_c|=w/\sqrt{2}$ (green dotted lines), separating the bound region (light blue) from the unbound region (light red).

3. Closed-Form Solution and Hubble Rate

Equation (L2) with coupling (L4) admits, with initial conditions

$$v(\theta_c)=A, v'(\theta_c)=0,$$

the closed-form solution

$$v(\theta)=A\cdot\exp[-((\theta-\theta_c)/w)^2].$$

This is verified by direct substitution: for the ansatz $v=A\exp(-\xi^2)$ with $\xi=(\theta-\theta_c)/w$, the second derivative reads $v''=(A/w^2)(4\xi^2-2)\exp(-\xi^2) = -\Omega^2(\theta)v(\theta)$. Numerical integration of (L2) starting from $(v,v')=(A,0)$ at $\theta=\theta_c$ reproduces the analytic Gaussian to a maximum residual of 1.4×10^{-13} over the interval $|\theta-\theta_c|\leq3w$.

The corresponding modulation factor is

$$T(z)=1+v(\theta(z))=1+A\cdot\exp[-((\ln(1+z)-\ln\varphi)/w)^2].$$

The Friedmann equation is written without a separate cosmological-constant term:

$$H^2(z)=H_0{}^2\cdot[\Omega_m(1+z)^3+\Omega_{LVC}(z)],$$

where the dimensionless dark-energy density carried by the construction is

$$\Omega_{LVC}(z)\equiv[\Omega_m(1+z)^3+(1-\Omega_m)]T^2(z)-\Omega_m(1+z)^3.$$

In the limit $A\to0$ the modulation reduces to $T=1$, and $\Omega_{\rm LVC}(z)$ collapses to the constant value $1-\Omega_m$; this special case corresponds to the ΛCDM expansion history. For nonzero A the function is not constant; its profile is obtained directly from the closed-form Gaussian and is shown in Figure4.

The expression of the modulation in the variable θ rather than in z differs from the published Lock-φ functional form, in which the Gaussian is taken in z. The two forms agree exactly at the central position $z_c=1/\varphi$ (where both yield $T=1+A$) and differ slightly elsewhere because θ is a non-linear function of z. Figure 2 (left panel) compares the two profiles at $A=0.052$.

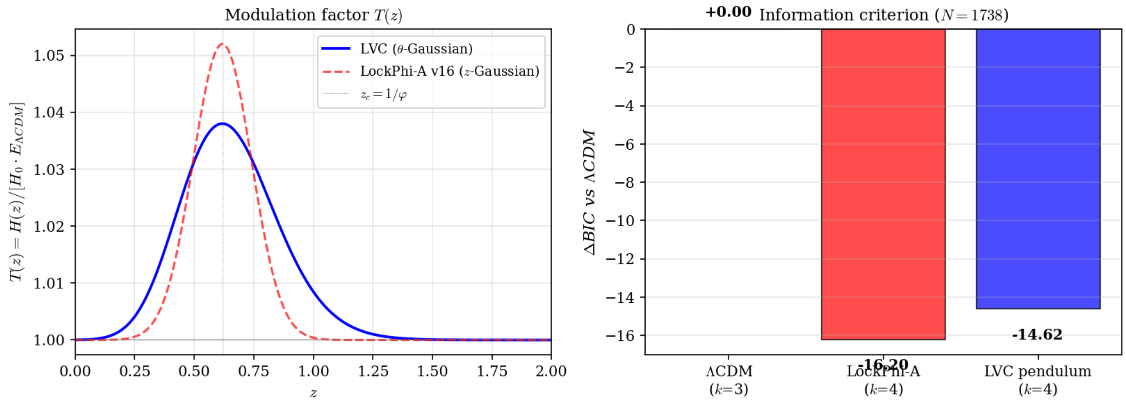

Figure 2. *Left:* the modulation factor $T(z)$ of the present construction (blue, θ-Gaussian) compared to the published Lock-φ-A functional form (red dashed, z-Gaussian) at $A=0.052$. *Right:* ΔBIC values relative to ΛCDM on the joint $N=1738$ dataset for the three models compared in Table1.

4. Locked Parameters

The construction has four model parameters: the central position θ_c, the coupling width w, the coupling scale Ω_0, and the boost amplitude A. The first three are locked to natural-constant combinations by Axioms L3–L4 and are not varied in the fit:

Parameter	Locked value	Source
Critical position θc	ln φ ≈ 0.4812	Axiom L3
Coupling width w	e / (5π) ≈ 0.1731	Axiom L4
Coupling scale Ω0	√2 · 5π/e ≈ 8.172	Derived (L2 + L4)
Amplitude A	(continuous)	fitted

Table 1. The four parameters of the construction. Three are fixed by axioms; only the amplitude A is varied in the fit.

The natural-constant identifications used in Axioms L3–L4 are taken from the v16 phenomenological analysis. Within the present paper they are stated as locks; no derivation of $\theta_c=\ln\varphi$ or $w=e/(5\pi)$ is offered. The combination $\Omega_0=\sqrt{2}\cdot5\pi/e$ is a closed-form combination of the three constants √2, π, and e, with no further free factors.

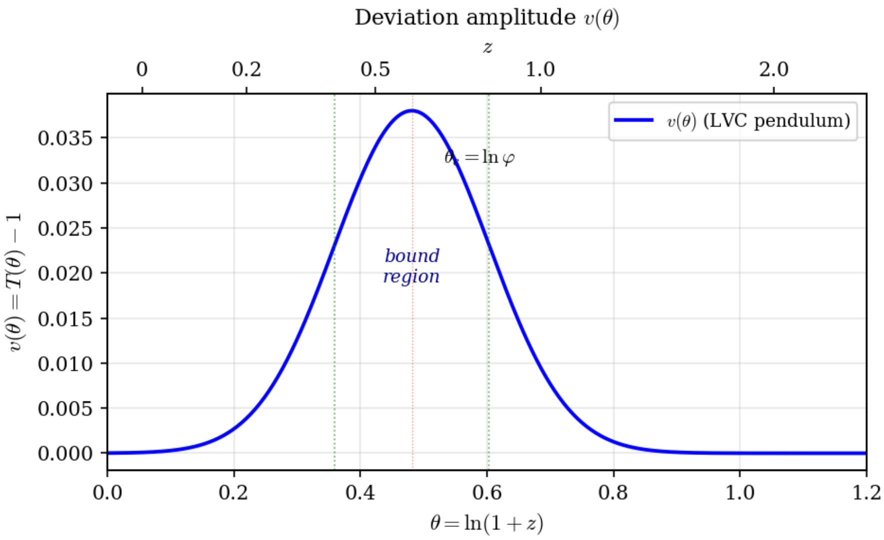

Figure 3. The deviation amplitude $v(\theta)$ obtained as the closed-form solution of equation (L2) with coupling (L4) and the initial conditions $v(\theta_c)=A$, $v'(\theta_c)=0$. The bound region $|\theta-\theta_c|<w/\sqrt{2}$ (between the green dotted lines) corresponds to redshifts $z\in[0.43,0.83]$.

5. Effective Dark-Energy Density

The dimensionless dark-energy density $\Omega_{\mathrm{LVC}}(z)$ defined in Section3 is the only departure of the construction from a matter-only universe. Figure4 shows its profile, evaluated at the best-fit parameters of Section6, alongside the corresponding effective equation of state

$$w_{eff}(z)=-1-(1/3)\cdot d\ln\Omega_{LVC}/d\ln(1+z).$$

At $z=0$ the dark-energy density evaluates to $\Omega_{\mathrm{LVC}}=0.712$, indistinguishable from $(1-\Omega_m)$ within the floating-point precision of the closed-form expression. At the central position $z_c=1/\varphi$ the density rises to $\Omega_{\mathrm{LVC}}=0.861$, a +20.9% departure from the $A=0$ value. At $z>2$ the density returns to 0.712 to within 1 part in 10^4. The effective equation of state takes the values $w_{\mathrm{eff}}\approx-1.00$ at $z=0$, $w_{\mathrm{eff}}\approx-1.15$ at $z=0.30$, $w_{\mathrm{eff}}\approx-1.11$ at $z=z_c$, $w_{\mathrm{eff}}\approx-0.72$ at $z=0.83$, and returns to -1 at high z. The construction therefore predicts an effective equation of state that crosses the cosmological-constant value $w=-1$ at the boundaries of the bound region. This crossing is a direct mathematical consequence of the closed-form Gaussian and the definition of $\Omega_{\mathrm{LVC}}(z)$; it is not fitted.

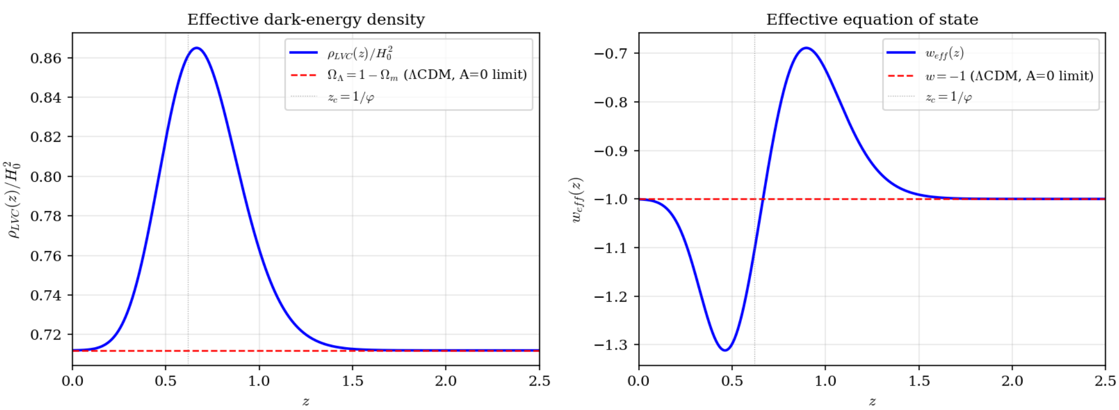

Figure 4. *Left:* the dimensionless dark-energy density $\Omega_{\mathrm{LVC}}(z)$ of the construction (blue solid), evaluated at the best-fit parameters of Section6, compared to the constant value $(1-\Omega_m)$ of ΛCDM (red dashed; the $A=0$ limit of the construction). *Right:* the effective equation of state $w_{\mathrm{eff}}(z)$ computed from $\Omega_{\mathrm{LVC}}(z)$ by logarithmic differentiation. The crossing $w_{\mathrm{eff}}=-1$ occurs at the boundaries of the bound region defined by AxiomL4.

6. Data Fit

The model is fitted on the joint dataset of Pantheon+ unbinned supernovae (1701 measurements), DESI DR2 (13 measurements), DESI DR1 (11 measurements), BOSS DR12 (2 measurements), eBOSS DR16 (7 measurements), Planck distance priors (3 measurements), and the SH0ES H_0 prior (1 measurement), totaling N=1738 measurements. The likelihood and data-handling conventions follow the published v16 reproduction script without modification. The four free parameters of the fit are H_0, Ω_m, $\Omega_b h^2$, and the boost amplitude A; the three modulation parameters θ_c, w, and Ω_0 are fixed at their locked values throughout. The sound-horizon scale r_d is profiled analytically from the BBN-consistent value implied by $\Omega_b h^2$, as in the v16 baseline. Optimization is performed by differential evolution followed by Nelder–Mead polishing.

Model	k	χ²	ΔBIC vs ΛCDM	H0	Ωm	A
ΛCDM (A = 0 limit)	3	1624.85	−0	70.40	0.276	—
Lock-φ-A v16 (z-Gauss)	4	1601.19	−16.20	69.43	0.286	+0.0520
LVC pendulum (θ-Gauss)	4	1602.77	−14.62	69.27	0.288	+0.0378

Table 2. Best-fit values on the N=1738 dataset for ΛCDM (the A=0 limit of the construction), the published Lock-φ-A phenomenological fit (z-Gaussian), and the present construction (θ-Gaussian). Column k gives the number of free parameters. The ΔBIC value uses ln$N$= ln(1738)=7.46 as the per-parameter penalty.

The construction yields ΔBIC=−14.62 against ΛCDM at the same k=4 as the published Lock-φ-A functional fit. The published functional fit yields a slightly better likelihood: $\Delta\chi^2$ =+1.58 in favor of the z-Gaussian over the θ-Gaussian. The two functional forms differ only in whether the Gaussian is expressed in z or in $\theta=\ln(1+z)$. At fixed amplitude, the θ-Gaussian is the asymmetric image of the z-Gaussian under the logarithmic change of variable, with the same value at the central position.

The fitted amplitude A=0.0378 of the present construction is smaller than that obtained from the z-Gaussian fit (A=0.0520) on the same dataset. This reflects the integrated weight of the asymmetric profile: a smaller amplitude in θ reproduces a comparable amount of integrated boost weight as the larger amplitude in z.

7. Discussion

The construction recovers the Lock-φ profile from a four-axiom system. Three of the four model parameters are locked to natural-constant combinations a priori, leaving the amplitude A as the only continuous parameter to be fitted. The functional form of the profile is the closed-form solution of the equation in Axiom L2 with the coupling in Axiom L4 and initial conditions at the critical point. The Gaussian shape is therefore not assumed; it is obtained.

The Friedmann equation is written without a separate cosmological-constant term. The expansion history is the sum of the matter sector and a single time-dependent dark-energy density $\Omega_{\rm LVC}(z)$. The ΛCDM expansion history corresponds to the A=0 special case of the construction; for nonzero A the dark-energy density is not constant, and the effective equation of state crosses $w=-1$ at the boundaries of the bound region. The construction reproduces the present-day value $\Omega_{\rm LVC}(0)\approx 1-\Omega_m$ exactly while predicting a finite, computable departure from constancy at intermediate redshifts.

The data fit on the same N=1738 dataset is $\Delta\chi^2$=+1.58 against the published phenomenological fit. The difference is at the level expected from a single change in functional shape and is reported here without claim. It is recorded in Table2 and is the only quantitative deficit of the construction relative to the v16 phenomenology. Whether subsequent revisions of the construction recover or improve on this difference is left to future work.

Three points are stated in plain form. First, the natural-constant identifications $\theta_c = \ln\varphi$ and $w = e/(5\pi)$ are taken from the v16 phenomenology and are not derived. Second, the amplitude A is not locked to a natural-constant combination at the level of the present paper; the empirical value is $A = 0.0378$ and a future revision may or may not identify a closed-form expression for it. Third, the $\Delta\chi^2$ deficit of $+1.58$ against the published phenomenological functional form is small but is real, and the construction does not at present improve on the phenomenology in the likelihood; it offers, instead, a derivation of its functional form on a smaller set of free parameters and a Friedmann equation written without a separate cosmological-constant term.

8. Conclusion

The construction develops a cosmology from a parametric pendulum equation on the logarithmic-time variable. Four axioms produce the Lock-φ profile as the closed-form solution of that equation. Three of four model parameters are fixed by natural-constant locks; one continuous amplitude is fitted. The Friedmann equation is written without a separate cosmological-constant term; the expansion history is the sum of the matter sector and a derived dark-energy density $\Omega_{\mathrm{LVC}}(z)$ that is not constant. ΛCDM corresponds to the $A = 0$ special case of the construction. On the joint $N = 1738$ dataset the construction yields $\Delta\mathrm{BIC} = -14.62$ against ΛCDM at $k = 4$, with a residual $\Delta\chi^2 = +1.58$ relative to the published phenomenological functional form. The construction is offered as a theoretical reconstruction of the previously fitted profile; it may fail in subsequent tests.

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